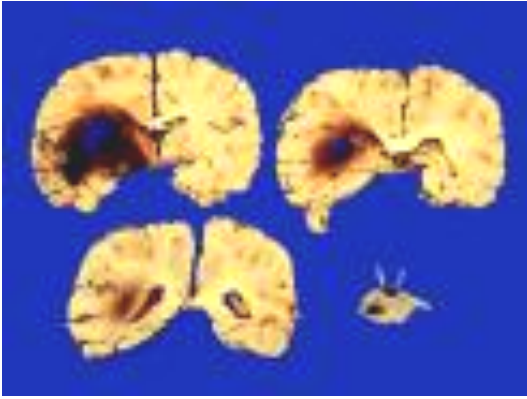
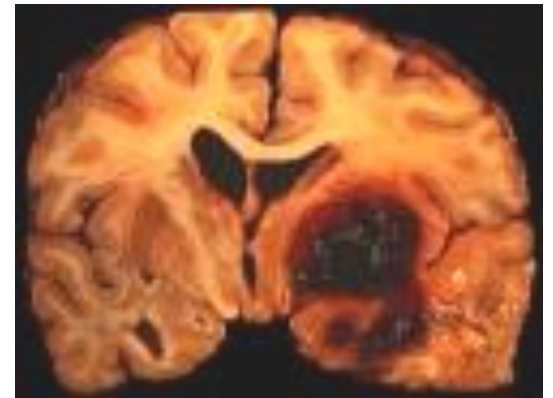




# Intracranial Hemorrhage



## Spontaneous, Nontraumatic Intracerebral Hemorrhage...

...occurs when a diseased blood vessel within the brain bursts, allowing blood to leak inside the brain.

# ICH Epidemiology

- 30 day mortality: 35-52%
  - 50% of these in first 48 hours
- One-fifth of survivors are independent at 6 months

# ICH Types

- Epidural
- Subdural
- Subarachnoid
- Intraparenchymal
- Intraventricular

# Intraparenchymal Bleeding

## Causes

### Primary:

- Hypertension (35%)
- Amyloid angiopathy (20%)
- Antiplatelet use and bleeding diathesis (5%)
- Drug/substance abuse

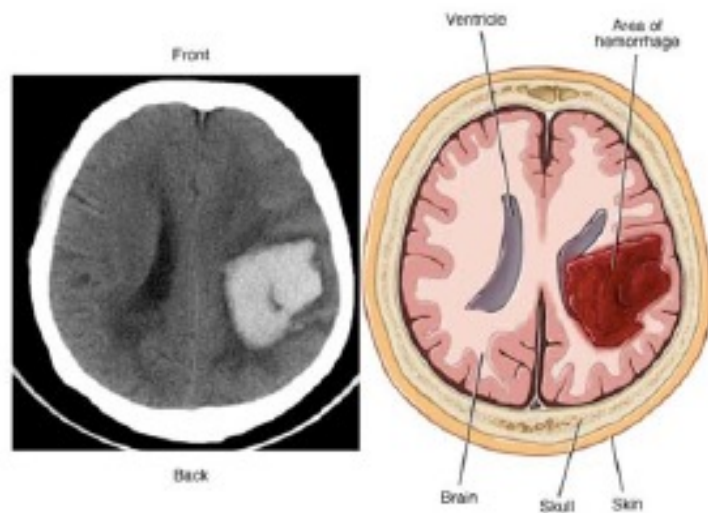
Cause remains undetermined in approximately 21% of spontaneous ICH.

### Secondary:

- Vascular Anomalies (5% overall but ~ 38% in ICH < 45 yrs of age):
  - Arterio-Venous malformation
  - Aneurysm
  - Cavernoma
  - AV fistula, etc.
- Anticoagulation (14%)
- Trauma
- Tumors
- Infection
- Hemorrhagic transformation of cerebral infarct
- Moyamoya disease
- Cerebral venous thrombosis

Meretoja, et al. Stroke, 43, 2012

# Highest Risk Factors

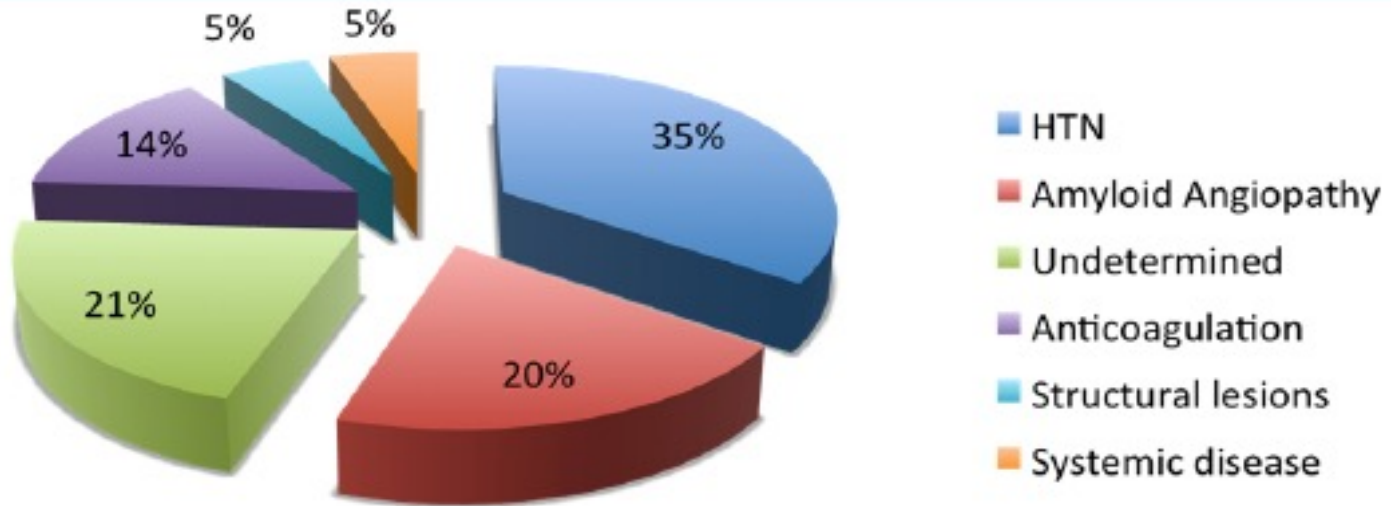


- Advanced age (>70)
- HTN
- Kidney Disease
- Heavy ETOH use
- Substance abuse (cocaine, amphetamines)
- Very low cholesterol

Juvela, et al. Stroke, 26, 1995.

Valappil, et al. Annals of Indian Academy of Neurology, 15(1), 2012.

# Causes of Spontaneous ICH



**HTN & Amyloid Angiopathy cause  
>50% of spontaneous ICH**

Meretoja, et al. Stroke, 43, 2012

# Hypertensive ICH

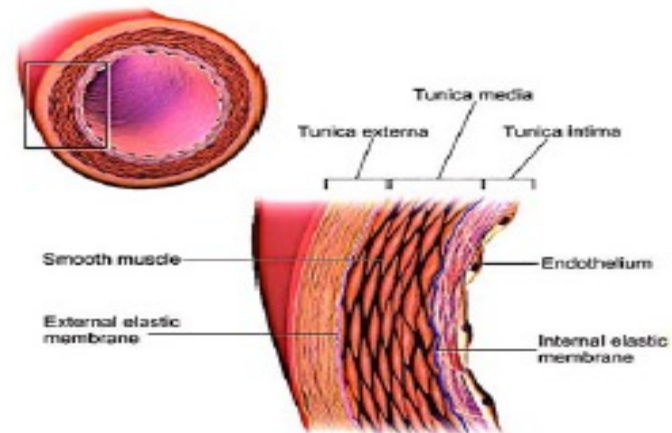
- Basal ganglia (50%)
  - Contralateral hemiparesis, sensory loss, conjugate gaze
- Lobar regions (20-50%)
  - Contralateral hemiparesis or sensory loss, aphasia, neglect, or confusion
- Thalamus (10-15%)
  - Contralateral hemiparesis, sensory loss, gaze paresis
- Pons (5-12%)
  - Quadriparesis, facial weakness, decreased level consciousness
- Cerebellum (1-5%)
  - Ataxia, miosis, gaze paresis



# Pathophysiology overview

## Hypertension induced ICH

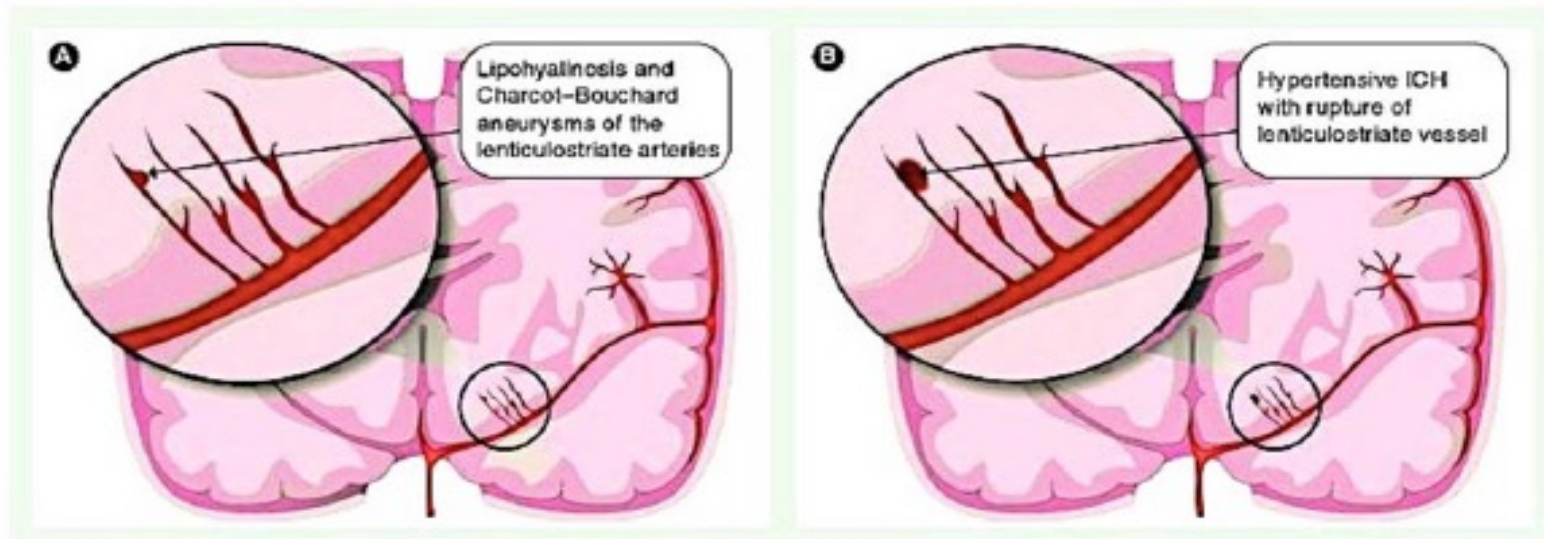
- Lipohyalinosis: Small vessel disease. Lipid accumulation in vessel wall causing narrowed lumen.
- Fibrinoid Necrosis: Accumulation of amorphous, basic, **proteinaceous** material in the tissue matrix with a staining pattern reminiscent of fibrin.  
**NOT FIBRIN!**
  - Antigen-Ab complexes stick to vessel wall -> attract inflammatory cells -> activate complement leading to cell wall damage and necrosis.
- Hyperplasia of the media in artery walls due to proliferation of reactive smooth muscle cells die -> cells replaced by collagen fibers -> brittle vessel wall



Chakrabarty, A., Shivane, A. 8(1), 2008

# Hypertension induced ICH:

- Charcot-Bouchard aneurysms: lenticulostriates, thalamoperforators, paramedian branches of the basilar artery, superior cerebellar arteries, and anterior inferior cerebellar arteries.



Chakrabarty, A., Shivane, A. 8(1), 2008

# ICH Presentation

- Hypertension (90%)
- Altered mental status (50%)
- Headache (40%)
- Seizures (6-7%)

# Intracerebral Hemorrhage Diagnosis

## CT Scan



Marc Dorfman, MD, FACEP, MACP

# Radiographical Imaging for ICH:

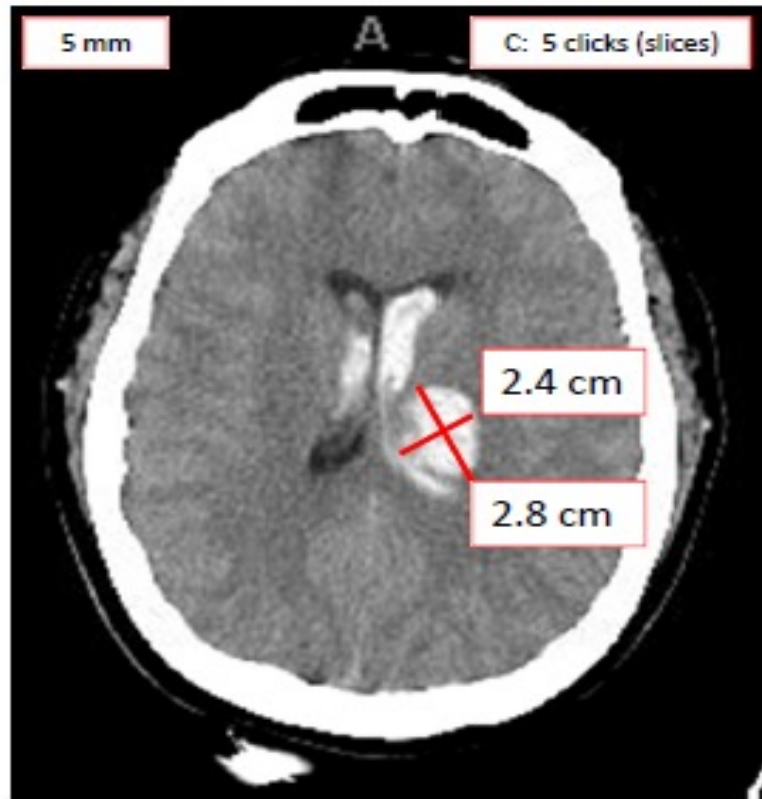
- Initial – CT scan
  - Repeat imaging for assessment of expansion should be CT for accurate comparison...  
“Apples to apples...”
- Diagnostic evaluation
  - Vascular lesions: Angiography
  - Tumors / Infections: MRI w/wo contrast
  - Trauma: CT good for blood but MRI best for prognostication



# Hemorrhage Assessment:

## Size of hemorrhage

A: Length (cm) x B: Width (cm) x C: Depth (# of slices) x slice thickness / 2



$$ABC \times 0.5 / 2$$

$$2.4 \text{ cm} \times 2.8 \text{ cm} \times 5 \times 0.5 / 2$$

$$= 8.4 \text{ ml ICH}$$

# Medical Management

| Correction of Coagulopathy |                          |                                           |                                                                             |
|----------------------------|--------------------------|-------------------------------------------|-----------------------------------------------------------------------------|
| Anticoagulant              | Reversal strategy        | Mechanism                                 | Approval status                                                             |
| Heparin, LMWH              | protamine                | Binding and inactivation of heparins      | FDA-approved                                                                |
| Warfarin                   | Vitamin K                | Overwhelm warfarin inhibition of VKOR-C1  | FDA-approved                                                                |
|                            | KCentra® (4-factor PCC)  | Factor replacement                        | FDA-approved                                                                |
| Dabigatran (Pradaxa®)      | Idarucizumab (Praxbind®) | Monoclonal antibody that binds dabigatran | FDA- approved                                                               |
| Rivaroxaban (Xarelto®)     | KCentra® (4-factor PCC)  | Factor replacement                        | Not-FDA approved<br>Use based on very low quality and inconsistent evidence |
| Apixaban (Eliquis®)        |                          | Decoy Xa molecule                         | Under FDA review                                                            |
| Edoxaban (Savaysa®)        |                          |                                           |                                                                             |

# Medical Management:

**Next most important...  
Blood Pressure Control**



# Blood Pressure Goals

What are we trying to achieve?

...Maximal cerebral perfusion of both global  
parenchyma AND local, perihematoma tissue  
being compressed...



## AHA / ASA Recommendation:

For ICH patients presenting with SBP between 150 and 220 mm Hg and without contraindication to acute BP treatment, acute lowering of **SBP to 140 mm Hg is safe** (Class I; Level of Evidence A) **and can be effective for improving functional outcome** (Class IIa; Level of Evidence B)

(Revised from the previous guideline)

For ICH patients presenting with SBP >220 mm Hg, it **may be reasonable to consider aggressive reduction of BP** with a continuous intravenous infusion and frequent BP monitoring.

(Class IIb; Level of Evidence C). (New recommendation)

# Glucose Management:

- High blood glucose on admission predicts increased risk of mortality and poor outcome
- Aggressive glucose management leads to increased incidence of both systemic and cerebral hypoglycemic events = increased risk of mortality

## AHA / ASA Recommendation:

Closely monitor glucose but avoid both hyperglycemia and hypoglycemia. Goal 140-180 mgdL.

(Class I; Level of Evidence C).

# Temperature Management:

- Fevers worsen outcome
- Increases cerebral metabolic demand
- Common in ICH, especially those with IVH, due to huge inflammatory response
- Hypothermia thought to reduce perihematomal edema but no evidence.
- Goal should be normothermia

## **AHA / ASA Recommendation:**

Treatment of fever after ICH is reasonable.

(Class IIb; Level of Evidence C). (New recommendation)

# Temperature Management:

## **Pharmacological Management:**

- Acetaminophen and/or NSAIDs
- Alternate, Dual antipyretics best!
- Watch for renal and/or hepatic toxicity
- With NSAIDs, watch for gastric irritation/ulcers and/or bleeding.

Mullins, et al. Neurocritical Care, 15(3), 2011

# Temperature Management:

## **Non-Pharmacological Interventions:**

- Cold fluid IV infusions
- Air circulating blankets
- Water-circulating blankets and pads
- Intravascular cooling catheters

## **Shiver Control:**

- Meperidine, buspirone, IV MgSO<sub>4</sub>
- Surface warming

Lopez, G.A. Curr Treat Options Neurol, 18:12, 2016

# Seizures and Seizure Prophylaxis:

## **AHA / ASA Recommendation:**

Clinical or subclinical seizures should be treated with anti-seizure drugs.

(Class I; Level of Evidence A)

Prophylactic anti-seizure medication is not recommended

(Class III; Level of Evidence B)

Consider continuous EEG when mental status disproportionate to degree of brain injury.

(Class IIa; Level of Evidence C)



# Seizures and Seizure Prophylaxis:

Neurocritical Neurocrit Care (2011) 15:80–84  
DOI 10.1007/s12028-010-9341-6

## ORIGINAL ARTICLE

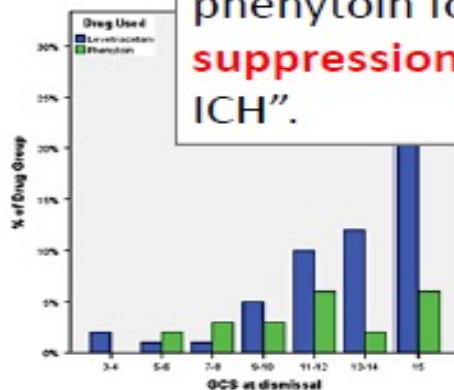
### Levetiracetam is Associated with Improved Cognitive Outcome for Patients with Intracranial Hemorrhage

Scott Taylor · Ro  
As'ad Ehtisham

## Retrospective analysis

N = 85  
(25-phenytoin vs  
levetiracetam)

Found “levetiracetam is **more effective** than phenytoin for seizure prophylaxis **without suppression of cognitive abilities** in patients with ICH”.



|                                   |            |          |      |
|-----------------------------------|------------|----------|------|
| Hospital length of stay           | 10.6 ± 8.4 | 12 ± 7.2 | 0.49 |
| Seizure incidence                 | 8%         | 0.0%     | 0.03 |
| NCCU discharge destination (home) | 16%        | 21.7%    | 0.04 |
| Cognitive function (intact)       | 36%        | 56.7%    | 0.08 |

# Management of complication

- 1. Raised ICP
- 2. Infection
- 3. DVT

# ICH Surgical Indications

- Cerebral Hm ~5cm or (50mm<sup>3</sup>) or Cerebellar hemorrhage > 3 cm (30mm<sup>3</sup>) who are deteriorating or with brain stem compression and hydrocephalus from ventricular obstruction
- Vascular malformation if lesion is surgically accessible and patient has chance for good outcome
- Young patients with a moderate or large lobar hemorrhage who are clinically deteriorating

# Non-Surgical ICH Pts

- Small Hemorrhages ( $10 \text{ cm}^3$ )
- Minimal neurological deficits
- GCS  $< 4$  (excluding cerebellar hemorrhage with brain stem compression)

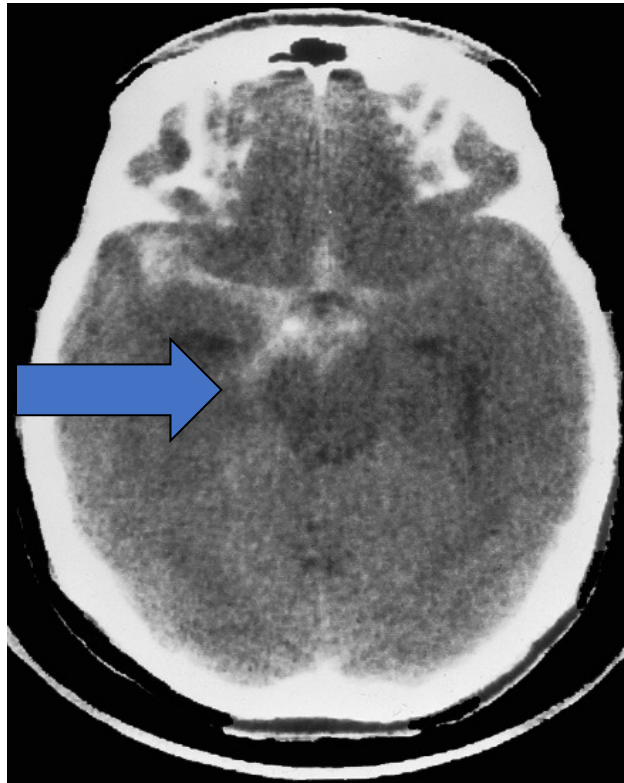
# Subarachnoid Hemorrhage

- Acute Evaluation
- “The worst headache of my life” is described by ~80% of patients
- “Sentinel” headache is described by ~20%
- Nausea/vomiting, stiff neck, loss of consciousness, or focal neurological deficits may occur

# Epidemiology

- SAH is ~1.6 times higher in women than men
- Risk factors for SAH include hypertension, smoking, female gender and heavy alcohol use
- Cocaine-related SAH occurs in younger patients
- Familial intracranial aneurysm (FIA) syndrome occurs when two first- through third-degree relatives have intracranial aneurysms

# CT Scan non-contrast showing blood in basal cisterns (SAH) – so called “Star-Sign”



CT Scan courtesy: University of Texas Health Science Center at San Antonio, Department of Neurosurgery

- The diagnostic sensitivity of CT scanning is not 100%, thus diagnostic lumbar puncture should be performed if the initial CT scan is negative



# Causes

- Aneurysm
- AVM
- Venous
- Others

# management

- 4 vessels angiography
- Prevent complication
- Rebleeding
- Vasospasm
- Seizure
- Electrolyte imbalance

- Pain killer
- Fluid therapy
- Nimodipine
- Laxative